

Thirulok Sundar Mohan Rasu

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Education

University of Michigan — Ann Arbor

Aug 2025 – Dec 2026

M.S., Electrical and Computer Engineering

GPA: 4.0 / 4.0

Relevant courses: *Algorithmic Robotics, Mobile Robotics, Advanced Multi-Robot Systems*

Indian Institute of Technology (ISM) Dhanbad

Dec 2021 – May 2025

B.Tech., Electrical Engineering. Member: *CyberLabs (Machine Learning)*

GPA: 8.30 / 10

Skills

Languages: Python, C++, MATLAB

Technical: Robotics Algorithms, Foundational models, Physics-based Modeling, Battery Management Systems, EIS Analysis, State-of-Health Estimation, Visuo-Tactile Sensing, Robot Manipulation, State Estimation, Motion Planning, SLAM, CV, ML

Tools: PyTorch, TensorFlow, ROS2, MuJoCo, Libero, Gazebo, PyBullet, OpenCV, Simulink, GitHub

Research Experience

HDR Lab — University of Michigan

Jan 2026 – Present

Graduate Research Assistant

- Developing computer vision techniques to enhance visuo-tactile sensor capabilities for robust object manipulation
- Built iOS app for automated visuo-tactile data collection enabling large-scale dataset generation for model training

Electric Vehicle Center — University of Michigan

Oct 2025 – Present

Graduate Research Assistant (Prof. Ziyu Song)

- Developed physics-based EIS modeling framework combining equivalent circuit models (SEI/CEI layers) with DRT analysis; extracted impedance parameters tracking SEI evolution from BOL to EOL across multiple cells, SOH, and SOC levels using Differential Evolution optimization
- Built attention-enhanced Seq2Seq models for battery aging prediction, achieving 2–5% MAPE on full capacity fade trajectories from first 30% of cycling data with robust generalization across C-rates and depths of discharge
- Designed accelerated testing protocol optimization framework converting real-world driving profiles into battery power demands via vehicle dynamics model; used Differential Evolution to select protocols maximizing degradation mechanism diversity (LAM-NE, LAM-PE, LLI-SEI, LLI-plating) for efficient lab testing

Carnegie Mellon University

Jan 2024 – Sep 2025

Research Intern (Dr. Arun Balajee Vasudevan)

- Developed benchmark audio deepfake dataset using multiple TTS models with paired real/synthetic audios; engineered end-to-end pipelines for generation, labeling, and evaluation (**paper**)
- Trained audio-language detection model combined with transcript-based LLM detection (Llama-8B) achieving state-of-the-art performance on synthetic audio identification

Mowito

Mar 2023 – Jul 2023

Robotics Software Intern

- Implemented perceptual hashing for training dataset deduplication, reducing dataset size by 30% while improving model training efficiency
- Built modular Dataset class supporting multiple annotation formats (instance, picking, crate) with visualization tools; integrated into production pipelines serving warehouse automation project
- Developed order-completion time estimation system using Leiden clustering (repo) for optimized crate grouping, reducing order fulfillment time

Selected Projects

OpenVLA model fine-tuning for robot manipulation (GitHub)

Fine-tuned OpenVLA (7B vision-language-action model) on LIBERO manipulation benchmarks using LoRA. Deployed on HPC cluster with NVIDIA A40 GPUs.

Robot State Estimation with Kalman & Particle Filters (GitHub)

Implemented and compared Extended Kalman Filter and Particle Filter for PR2 robot localization across five scenarios with complex trajectories; analyzed performance trade-offs between computational efficiency and estimation accuracy

Autonomous Mobile Robot Navigation System (GitHub)

Developed complete autonomous navigation stack for differential-drive robot using ROS2 and Nav2; implemented real-time obstacle detection and collision avoidance in C++ processing lidar data at 10Hz for dynamic environments